

Wi-Fi-enabled surveillance RC Car using ESP32 S3 Sense and Web Server

K explore (Robotics)

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OBJECTIVE

The aim of this design experiment is to build a Wi-Fi controlled RC car using an ESP32 microcontroller and a mobile application developed in MIT App Inventor. The project demonstrates how embedded systems integrate hardware and software to achieve real-time control, while also showcasing the use of wireless communication for flexible and remote operation.

THEORY

This project leverages the ESP32's Access Point (AP) mode to create a self-contained wireless ecosystem, integrating real-time camera streaming with precise motor control. The camera module directly interfaces with the ESP32, streaming live video through the onboard web server while simultaneously processing motion commands. This dual functionality enables operators to both control movement and receive visual feedback through a single interface, all without relying on external network infrastructure. The integrated system provides a complete remote inspection solution, particularly valuable in areas lacking Wi-Fi coverage, where immediate visual monitoring and mobility are required simultaneously.

REQUIREMENTS

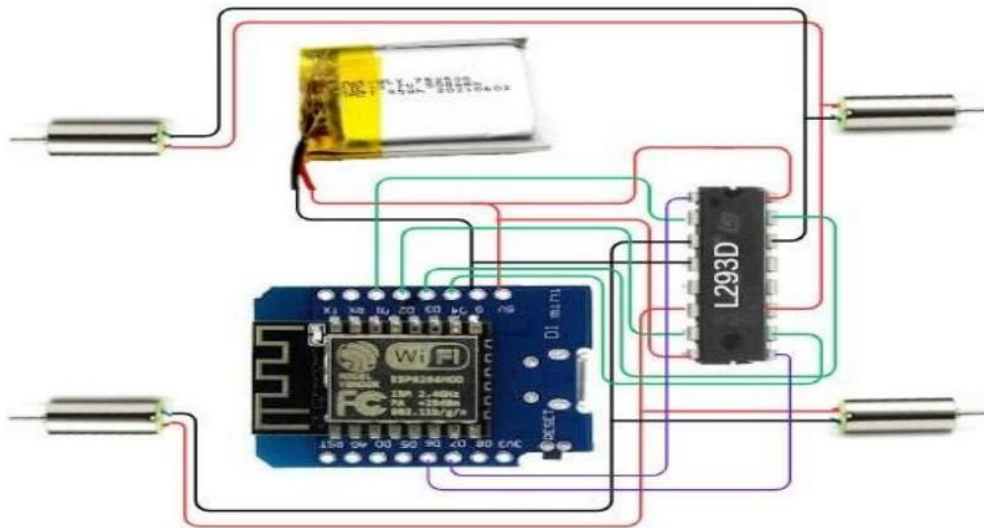
Hardware Requirements:

Components	Quantity
ESP32 S3 sense	1
L298N Motor Driver	1
DC Coreless Motor	4
Li-ion Battery	2
LEDs	2
Connecting Wires	As per required

Software Requirement:

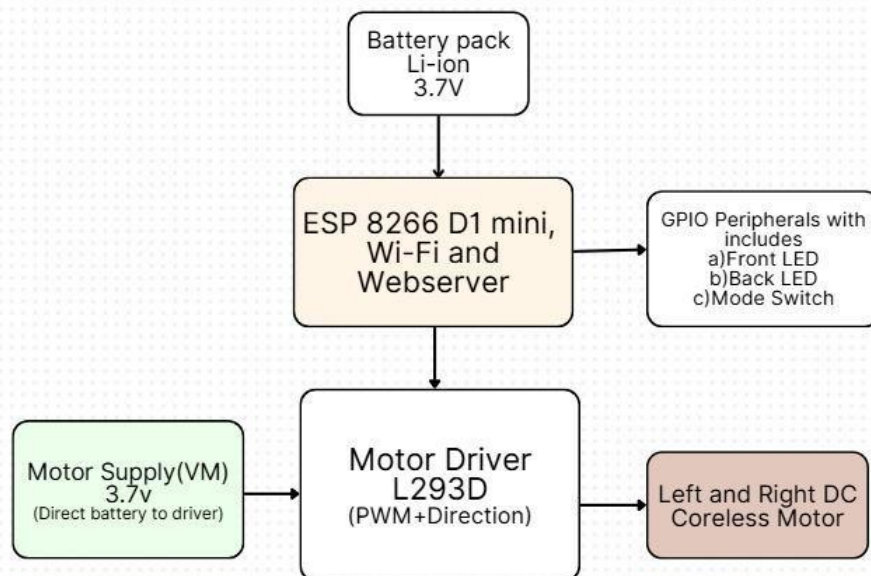
- a) Arduino IDE with ESP32 Board Packages
- b) MIT App Inventor

CIRCUIT DIAGRAM



FLOWCHART

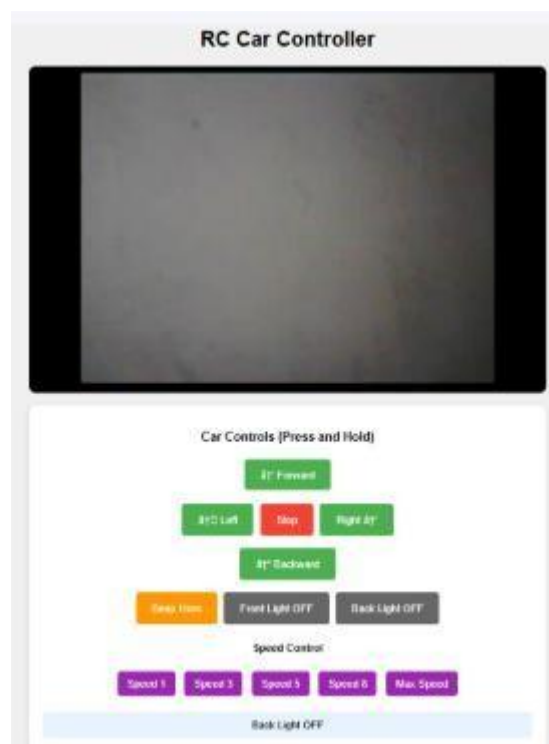
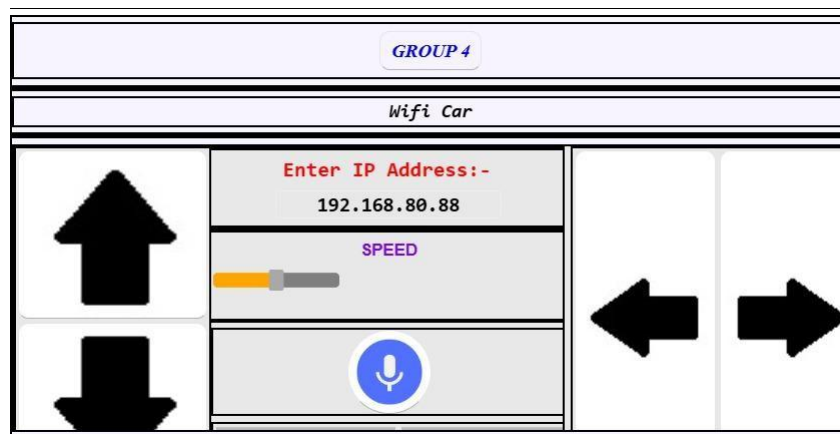
Hardware Flowchart - Wi-Fi RC Car

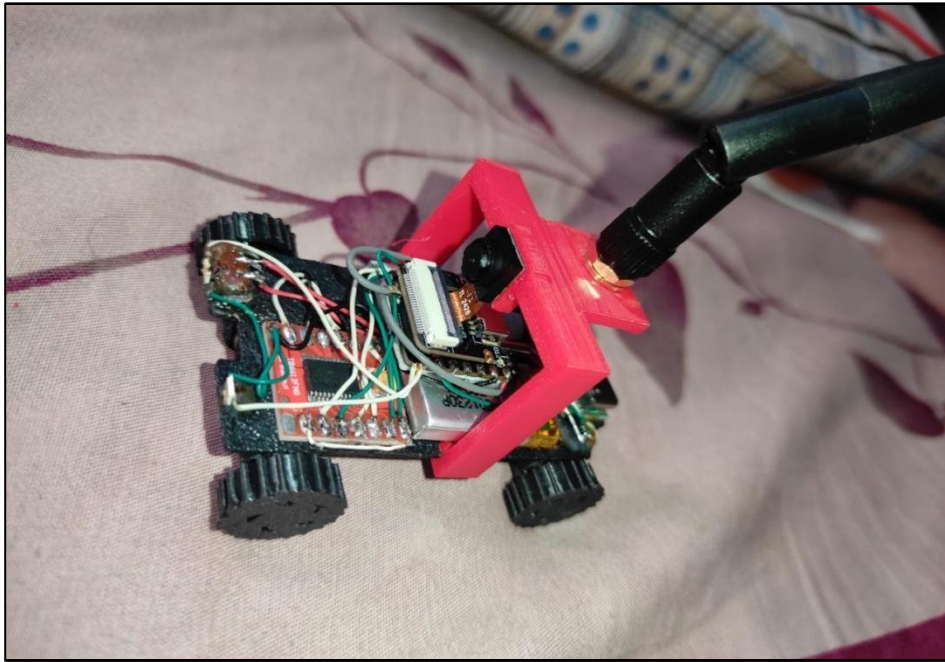


CODE DEVELOPMENT

This block diagram from MIT App Inventor represents the mobile app logic for controlling the Wi-Fi RC car. Each button (Forward, Backward, Left, Right) uses TouchDown to send a motion command and TouchUp to stop the car, ensuring safe control. Additional blocks handle headlights, backlights, hand speed adjustment through a slider. A voice command block is also included, enabling the car to respond to spoken instructions for flexible, hands-free operation.

User Interface in MIT App Inventor and WEB PAGE





Code Blocks Used to Control the RC Car



OBSERVATIONS

In this experiment, the designed Wi-Fi RC car was successfully tested under both Access Point (AP) mode and Station (STA) mode. The mobile application developed in MIT App Inventor was able to send commands to the ESP32 microcontroller, which then controlled the motors and lights through the motor driver circuit. The results were observed in terms of system responsiveness, speed control, and range of communication.

REAL-TIME APPLICATION/INDUSTRIAL SIGNIFICANCE

This project has several real-time applications and industrial significance:

Mining Industry – Wi-Fi controlled robotic vehicles can be deployed in underground mines to remotely inspect hazardous zones, monitor gas levels, or transport small loads, reducing risks to human workers.

Warehouse Automation – Similar systems can be adapted for automated guided vehicles (AGVs) in warehouses to move goods, perform inventory checks, or deliver items efficiently.

Search and Rescue Operations – A compact Wi-Fi rover with added sensors or cameras can navigate through collapsed structures or disaster zones to locate survivors where it is unsafe for humans to enter.

CONCLUSION

In conclusion, the Wi-Fi controlled RC car using ESP32 and MIT App Inventor successfully demonstrates the integration of hardware and software for real-time wireless control. The system achieved reliable motion, speed regulation, and auxiliary functions like lights while highlighting practical challenges such as latency and power limitations. Overall, the project validates the effectiveness of microcontroller-based design in developing cost-efficient, flexible, and application-oriented robotic systems.

[Drive link for video](#)

QR CODE FOR OUR PROJECT VIDEO

